
HS 550P: Statistical Methods Practical

Course Code	HS 550P
Course Name	Statistical Methods Practical
L-T-P-C	0-0-2-1
Prerequisites	This course should be taken along with HS550: Statistical Methods
Intended for	Postgraduate and B.Tech.
Distribution	HSS Course - Elective
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)
Discontinued	27th BoA (Nov 2018)

Course Contents

Representation of Data and Descriptive Statistics (2 hrs) Representation of data; Central tendency, dispersion, skewness, and kurtosis; Quartile and percentile; Gini coefficient and Lorenz curve; Correlation.

Random Sampling and Parametric Statistical Inference (6 hrs) Drawing random samples from the population - simple random sampling (with and without replacement); Sampling distribution; Statistical estimation - parametric point estimation and interval estimation; Maximum Likelihood Estimator, Hypothesis testing and calculation of effect size.

Non-parametric Statistical Inference (6 hrs) Carrying out non-parametric tests, estimation of location and dispersion, tolerance interval; non parametric measures and tests of association.

Designs of Experiment (6 hrs) Experimental design strategies; Blocking and Randomization; Factorial design of experiments.

Regression Analysis and ANOVA (8 hrs) Multiple regression analysis, interpretation of estimates, testing of significance; Analysis of

Variance (ANOVA) - one way ANOVA and two way ANOVA.

Text Books

1. Heiman, G. W., **Basic Statistics for the Behavioral Sciences**, 6th Edition, Wadsworth, 2011.

References

1. Field, A. (2013). *Discovering Statistics Using IBM SPSS Statistics*. Sage Publishing.
2. Field, A., Miles, J. and Field, Z., *Discovering Statistics Using R*, Sage Publishing, 2012.
3. STATA Manual: <https://www.stata.com/manuals13/u.pdf>